GPIE

Green Policy Intelligence Engine

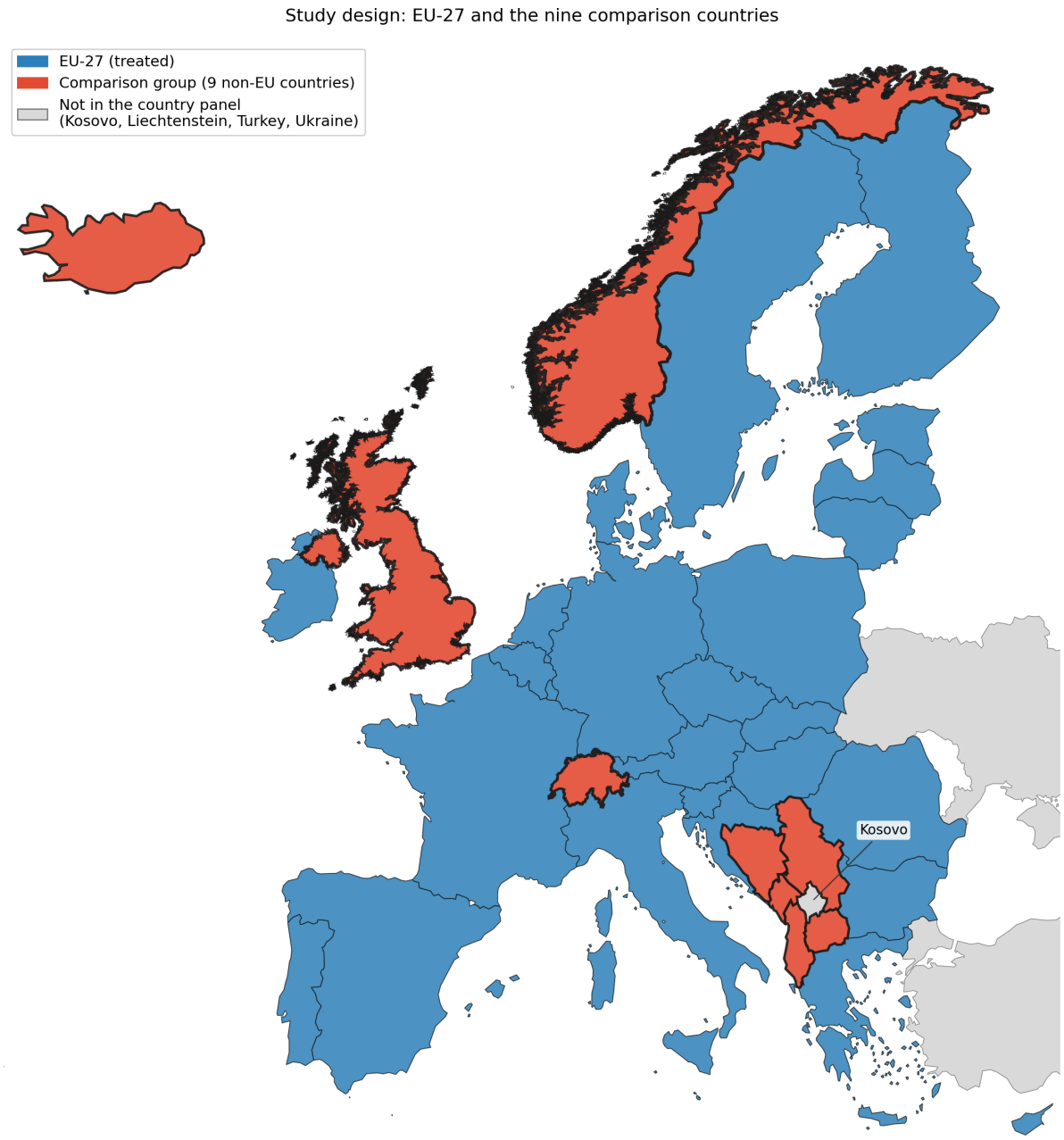
All Maps and Plots

24 figures compiled from outputs/plots/

1. Study Design: Treatment vs. Control Group
2. NO2 Choropleth Map
3. NO2 Before vs. After the European Climate Law
4. NO2: EU-27 vs. Control Group
5. Event-Study: Quarter-by-Quarter NO2 Effect
6. NDVI Choropleth Map
7. NDVI Before vs. After the European Climate Law
8. NDVI: EU-27 vs. Control Group
9. GDP Choropleth Map
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11. Elevation (DEM) Map
12. Climate / Temperature Map
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20. Pixel-Level NO2 Change Map
21. Local Moran's I (LISA) Spatial Cluster Map
22. Policies by Year
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24. Policy Types by Year

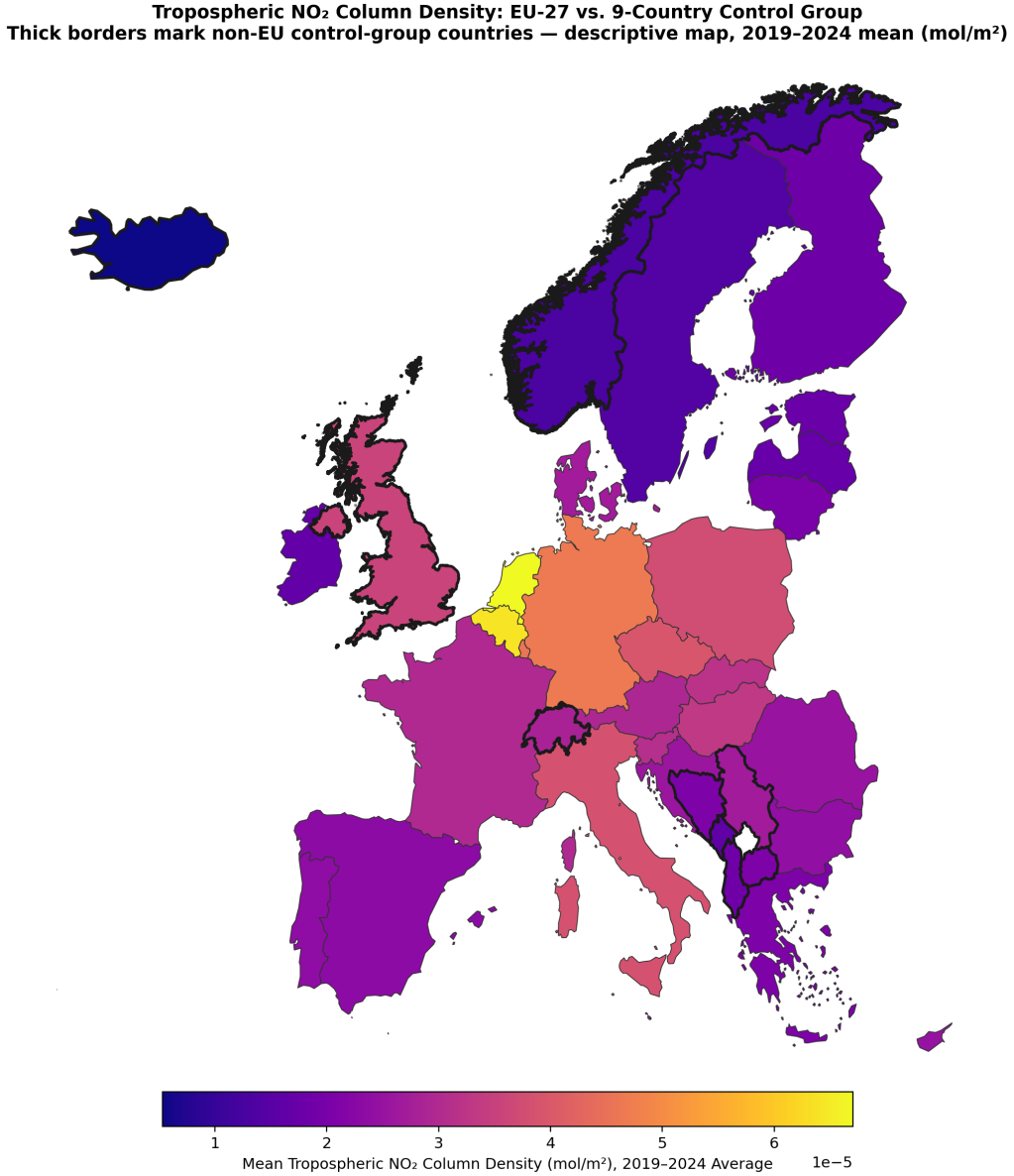
Study Design: Treatment vs. Control Group

he 9-country non-EU control group - UK, Norway, Switzerland, Iceland, Albania, Bosnia and Herzegovina, Montenegro, North Macedonia, Serbia. Kosovo, Liechtenstein, Turkey and Ukraine (grey) are



NO2 Choropleth Map

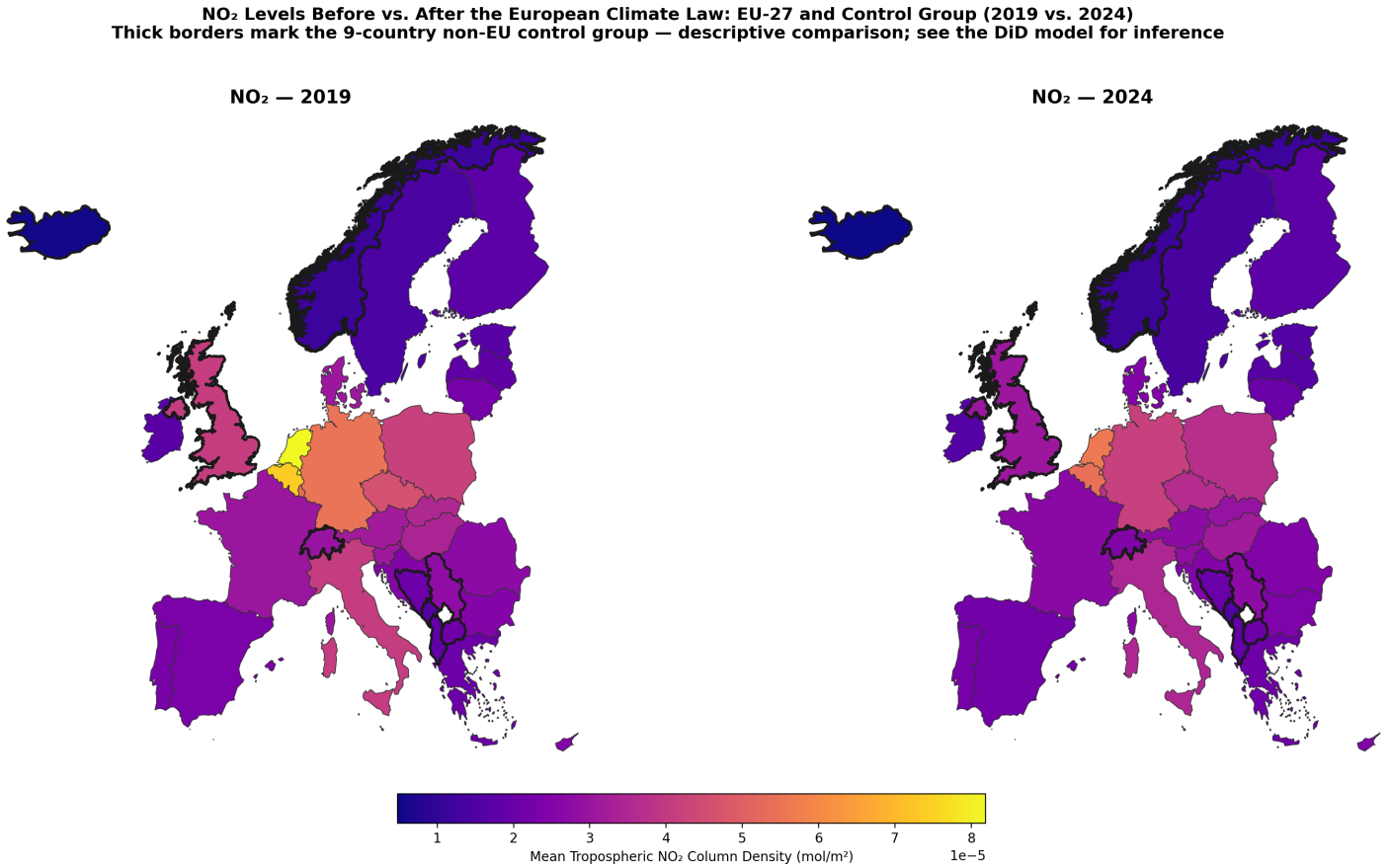
Mean tropospheric NO2 concentration by country, Sentinel-5P TROPOMI.



Green Policy Intelligence Engine (GPIE) — Source: Sentinel-5P TROPOMI, Sentinel Hub Process API

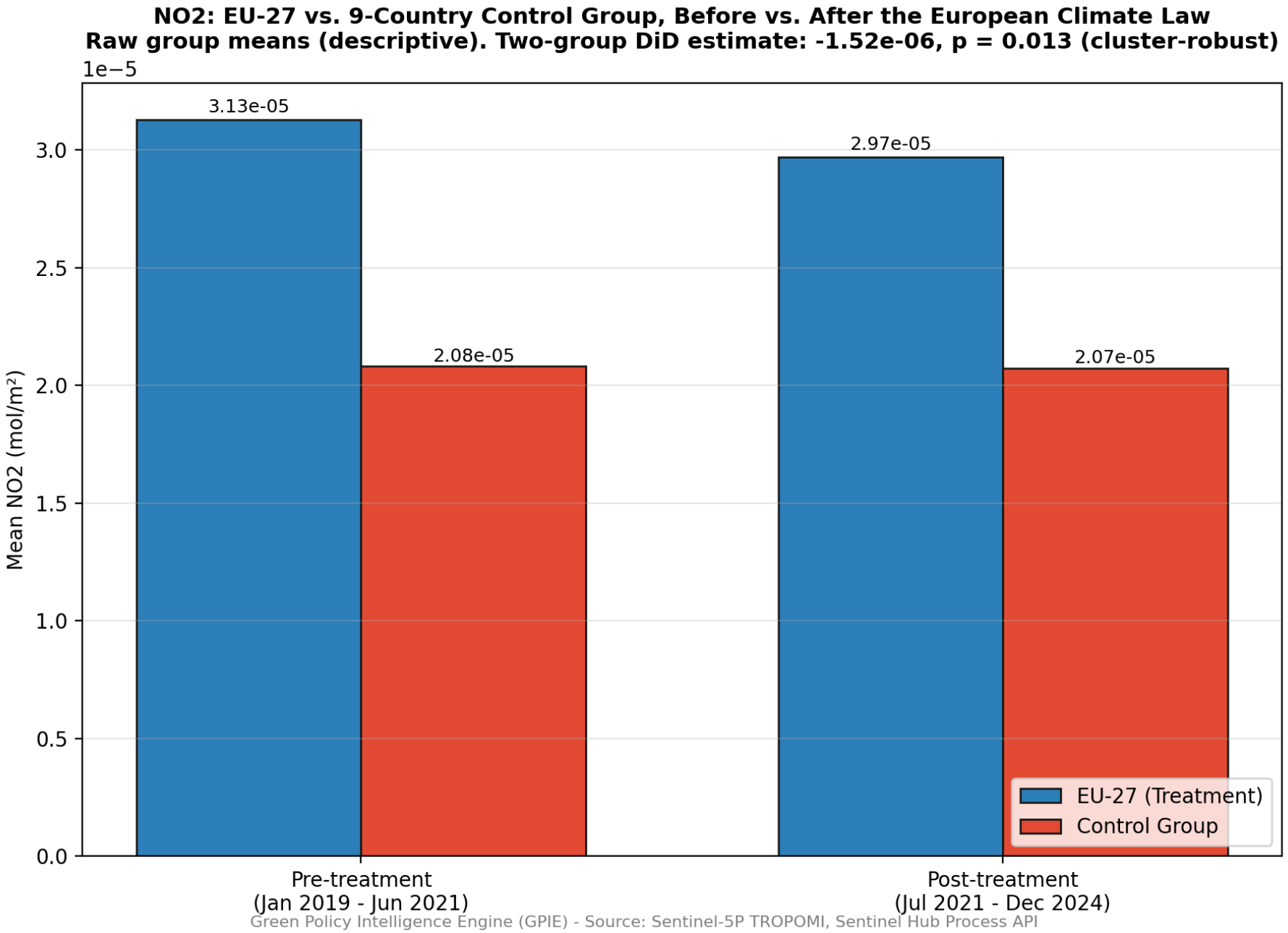
NO2 Before vs. After the European Climate Law

Country-level annual mean NO2, all 36 countries, 2019 (pre-treatment) vs. 2024 (post-treatment). Descriptive only.



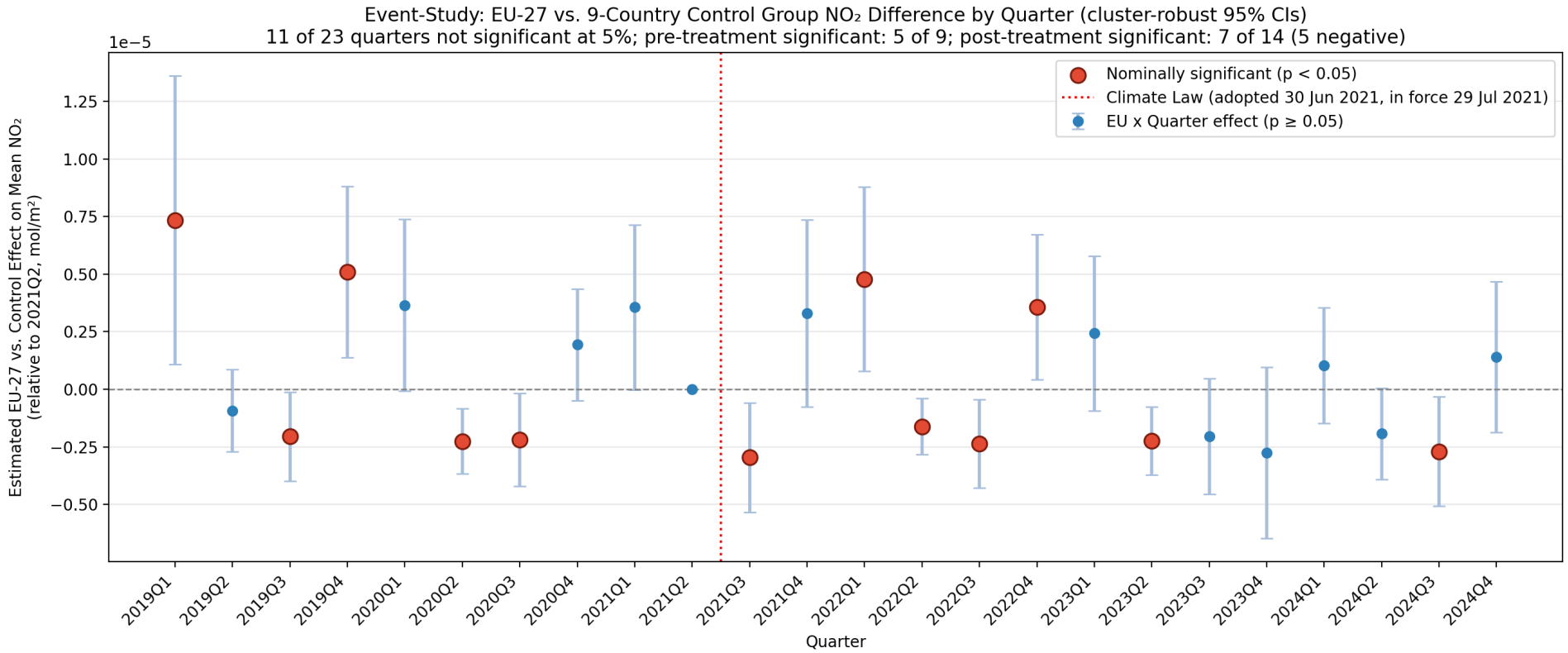
NO2: EU-27 vs. Control Group

Raw mean NO2, EU-27 vs. 9-country control group, Jan 2019-Jun 2021 vs. Jul 2021-Dec 2024 (descriptive; DiD estimate in title).



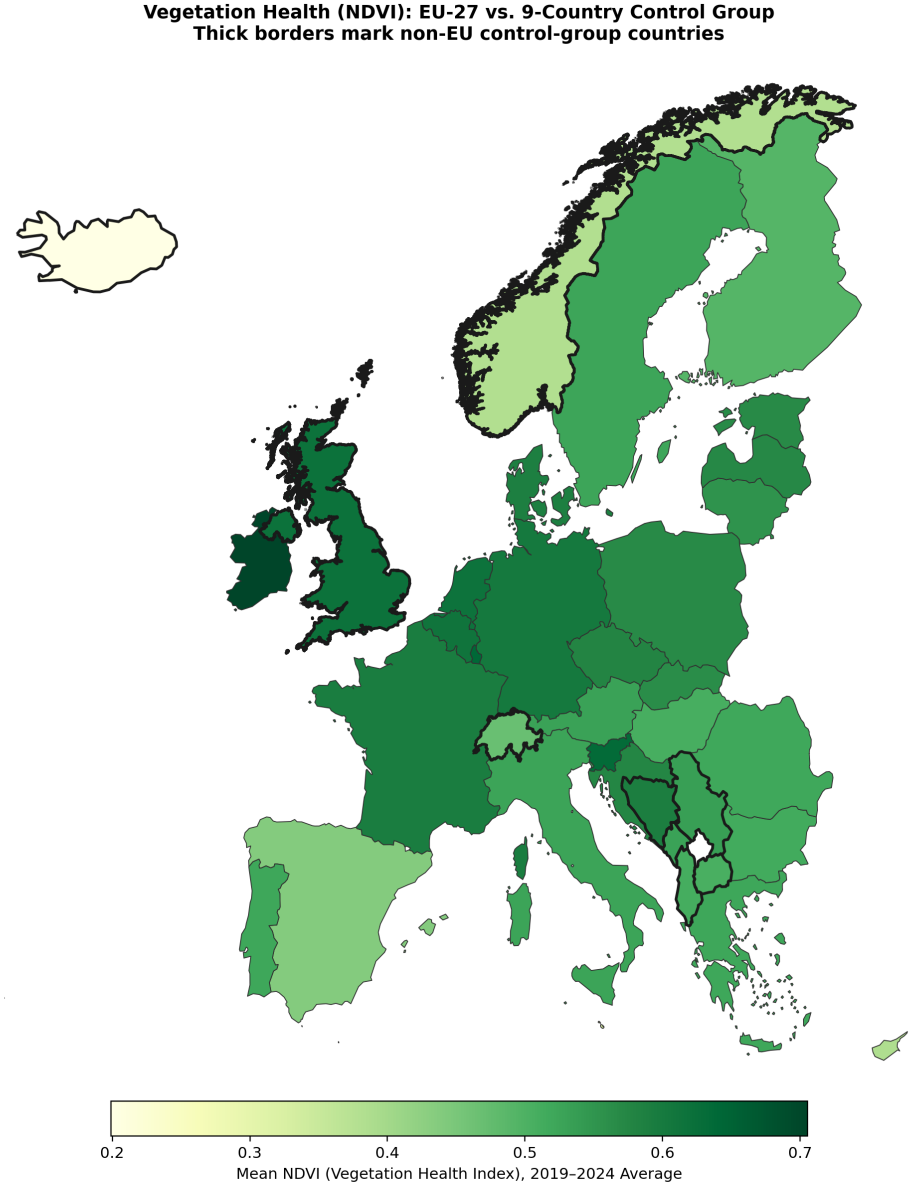
Event-Study: Quarter-by-Quarter NO2 Effect

23 quarter-specific EU x quarter coefficients relative to 2021Q2, cluster-robust 95% CIs (counts of significant quarters in the figure title).



NDVI Choropleth Map

Mean vegetation health index (NDVI) by country.

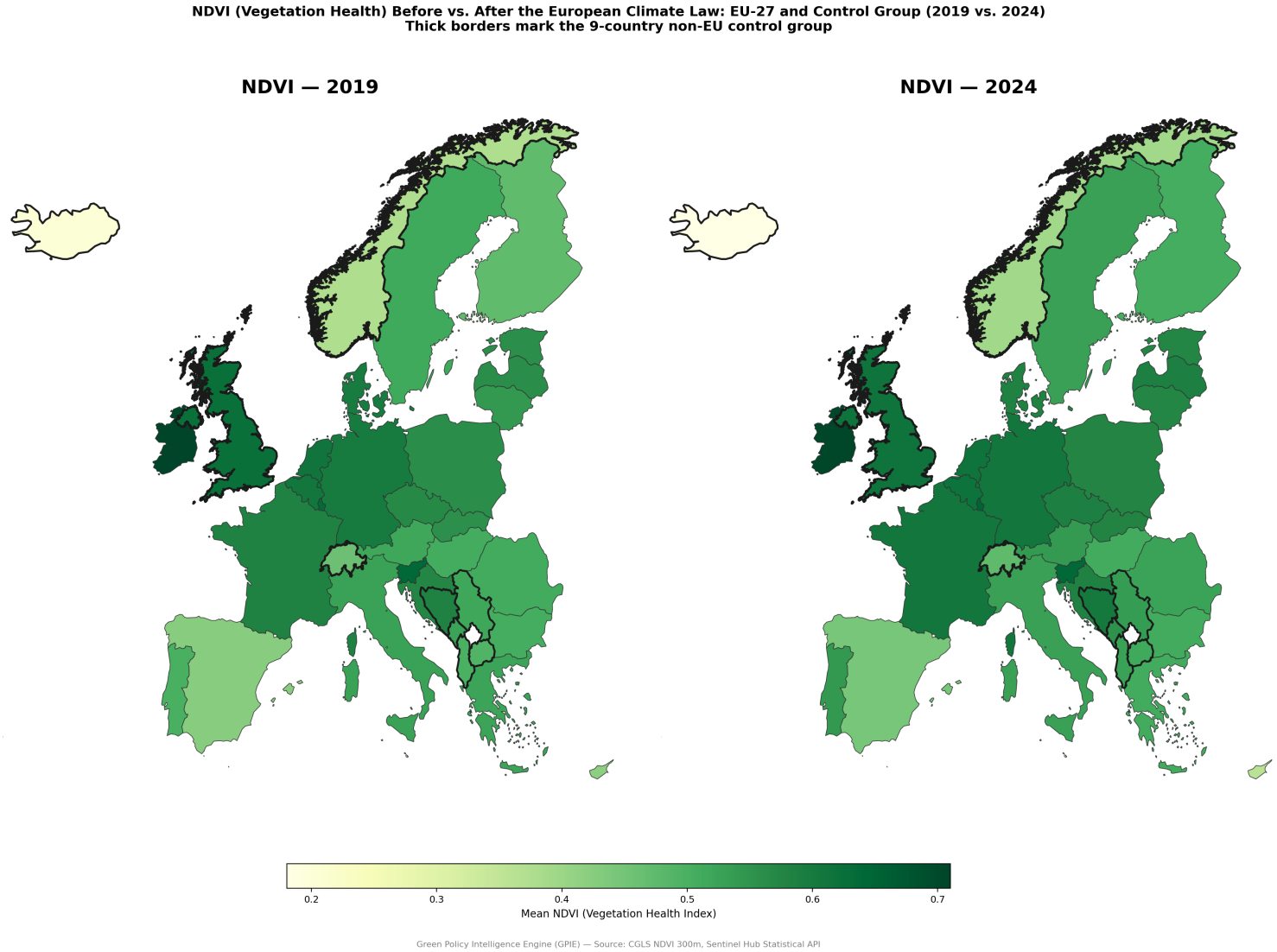


Green Policy Intelligence Engine (GPIE) — Source: CGLS NDVI 300m, Sentinel Hub Statistical API

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NDVI Before vs. After the European Climate Law

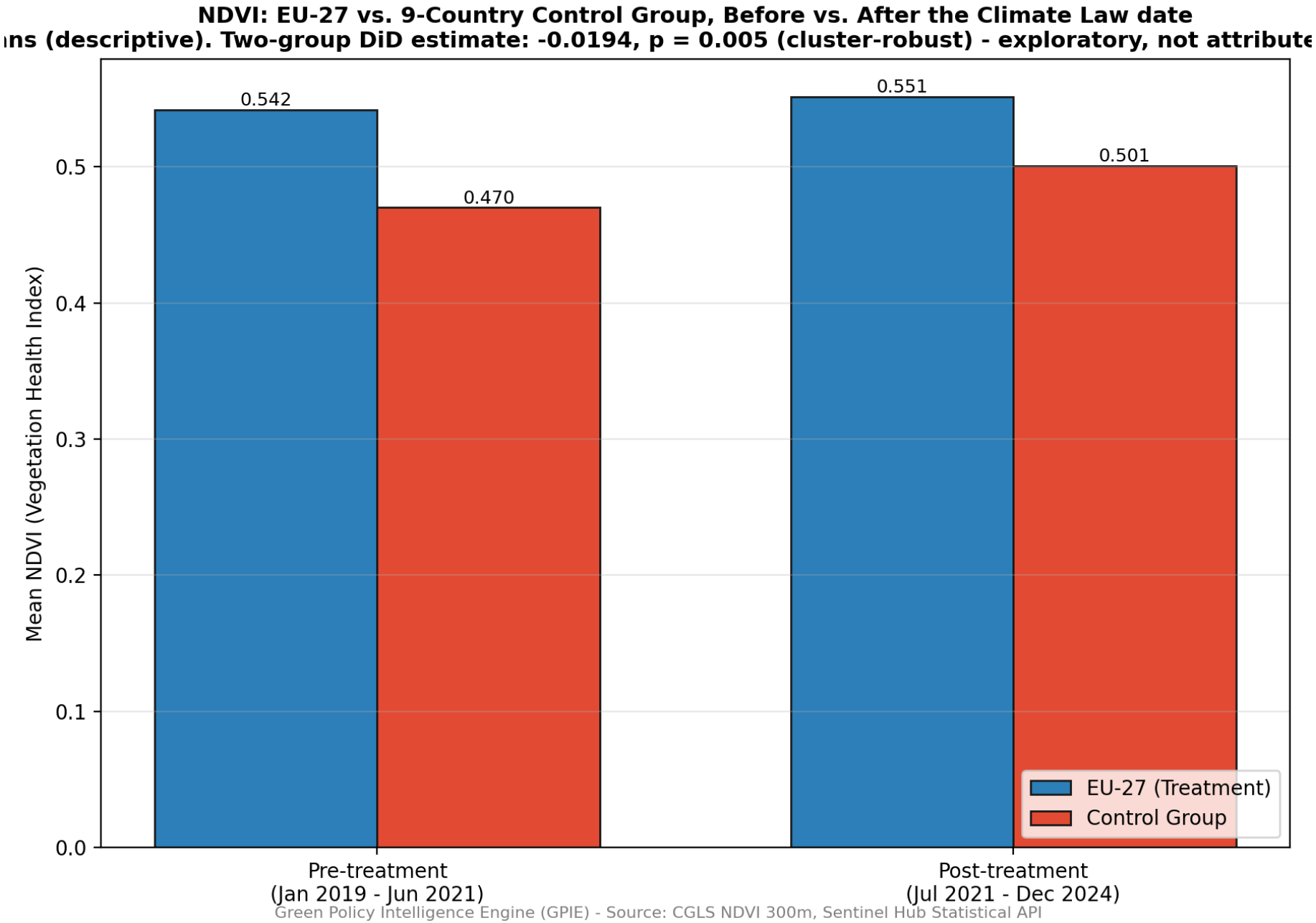
Country-level annual mean NDVI, all 36 countries, 2019 vs. 2024. Descriptive only.



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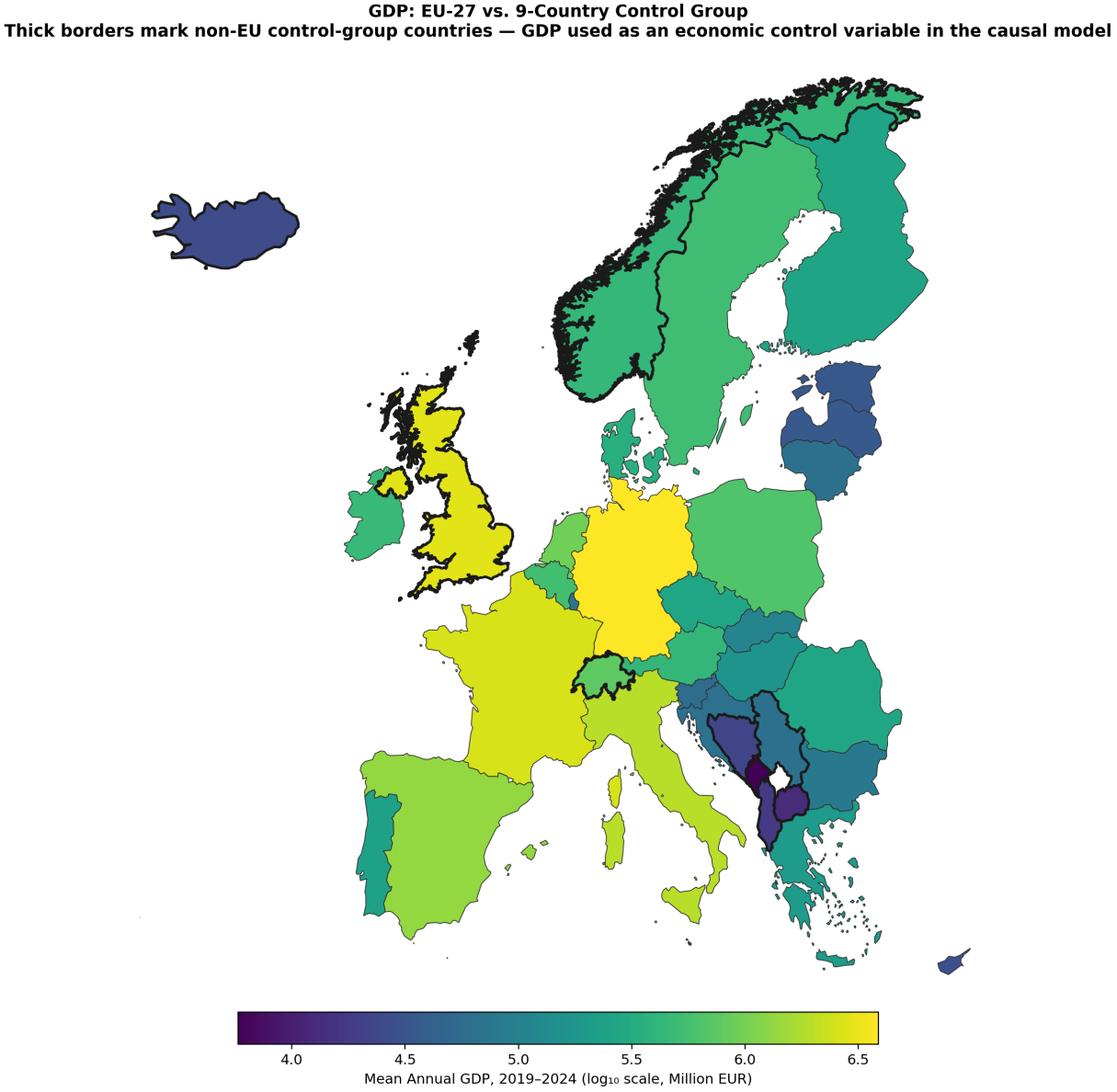
NDVI: EU-27 vs. Control Group

Raw mean NDVI, EU-27 vs. control group, pre vs. post (descriptive); exploratory DiD estimate in title, not attributed to the Climate Law.



GDP Choropleth Map

Control variable - GDP by country (annual, repeated across months in the panel).

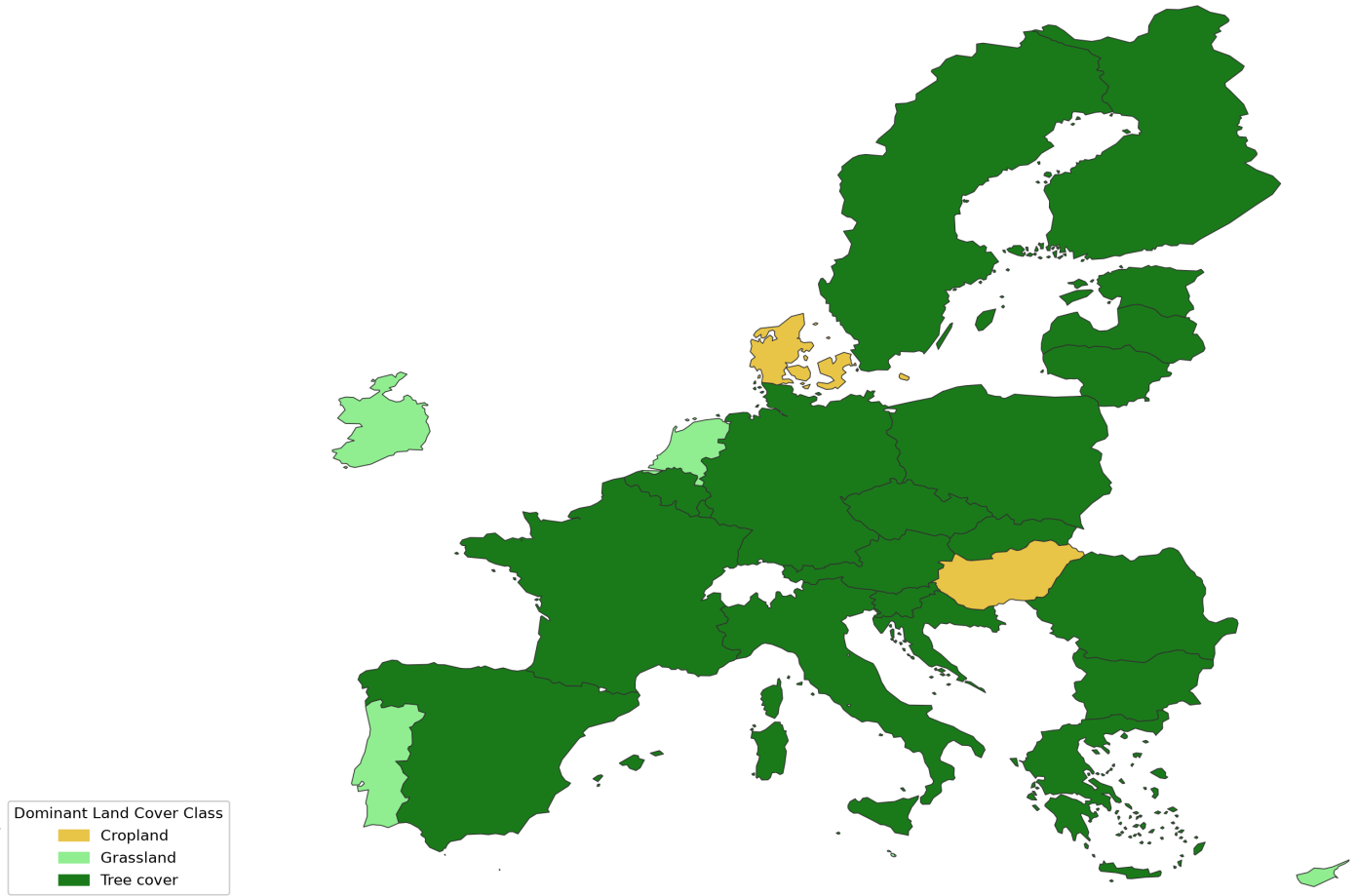


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Dominant Land Cover Class Map

Descriptive context - dominant land-cover class (static; absorbed by country fixed effects, not a model regressor).

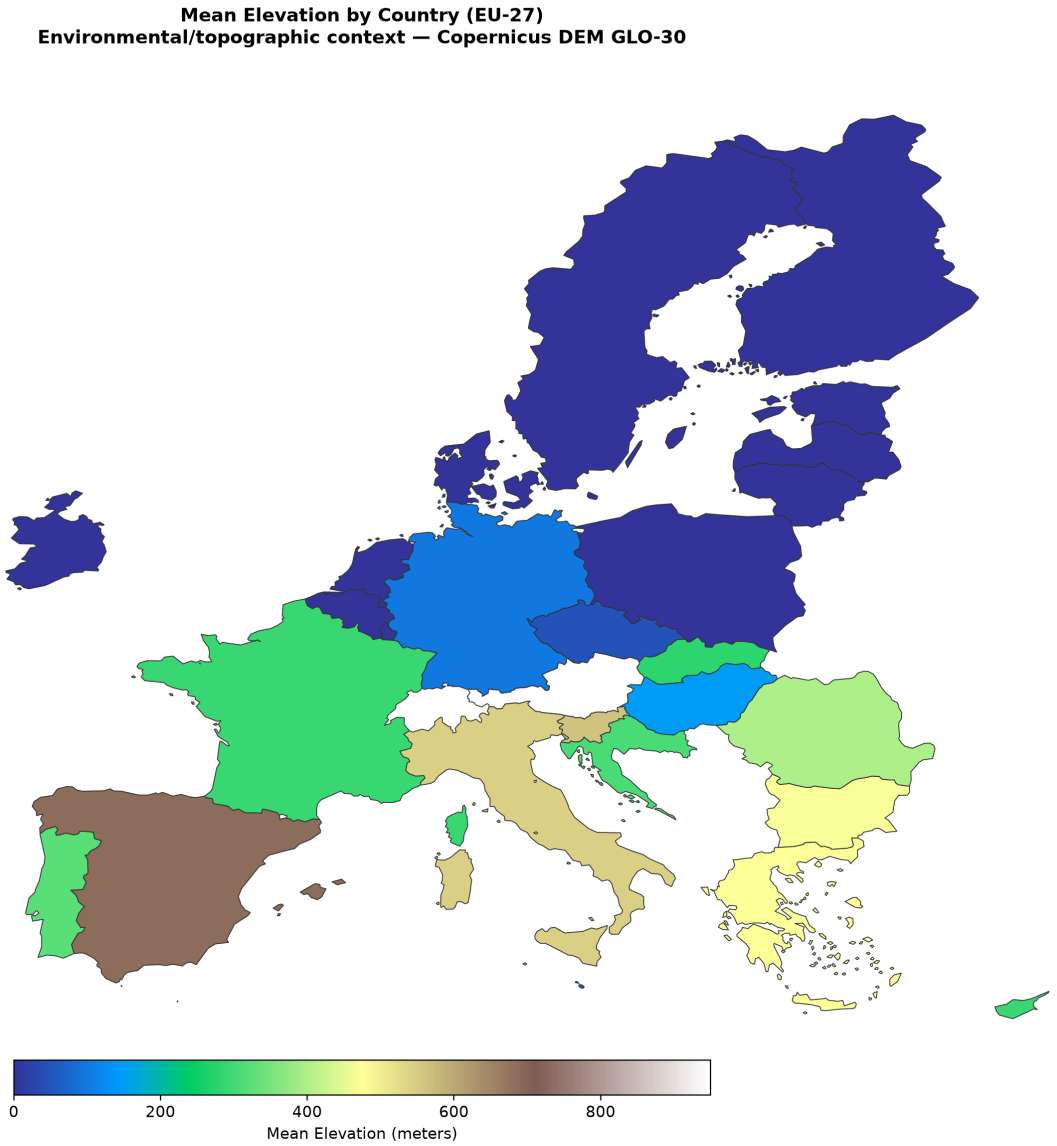
Dominant Land Cover Class by Country (EU-27)
Each country shaded by its single largest ESA WorldCover class (2021)



Green Policy Intelligence Engine (GPIE) — Source: ESA WorldCover 10m v200

Elevation (DEM) Map

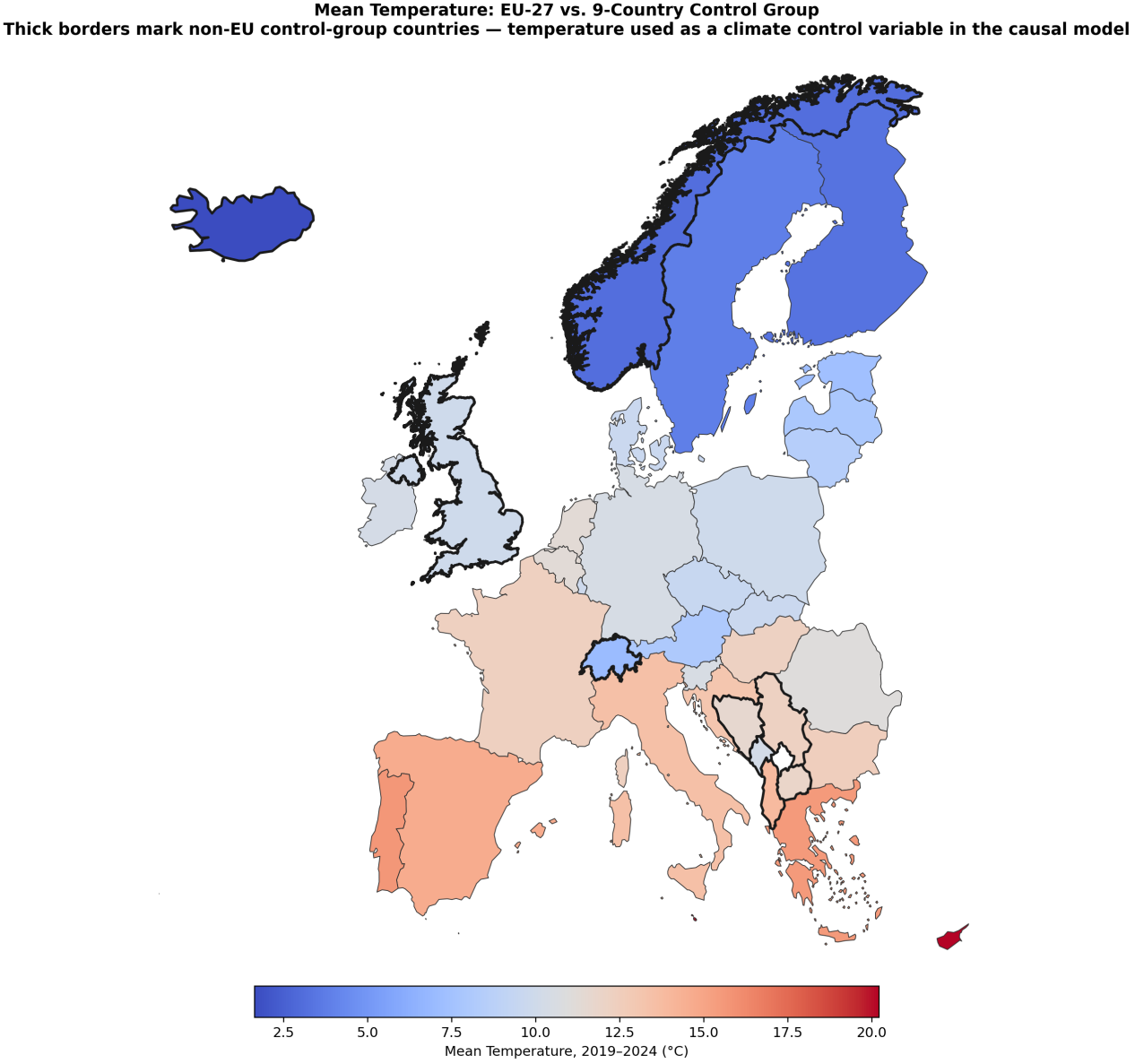
Descriptive context - mean elevation (static; absorbed by country fixed effects, not a model regressor).



Green Policy Intelligence Engine (GPIE) — Source: Copernicus DEM GLO-30

Climate / Temperature Map

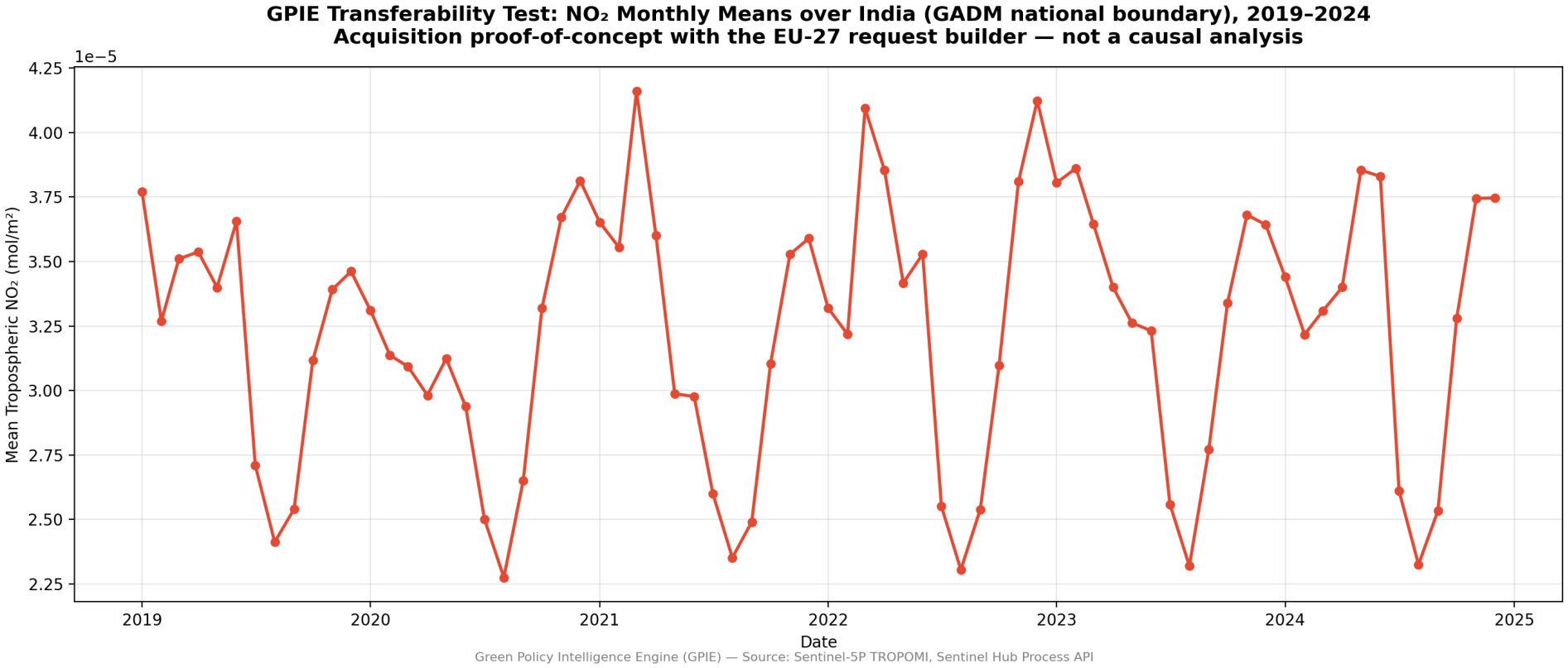
Control variable - average temperature by country.



Green Policy Intelligence Engine (GPIE) — Source: ERA5 Reanalysis, Copernicus Climate Data Store

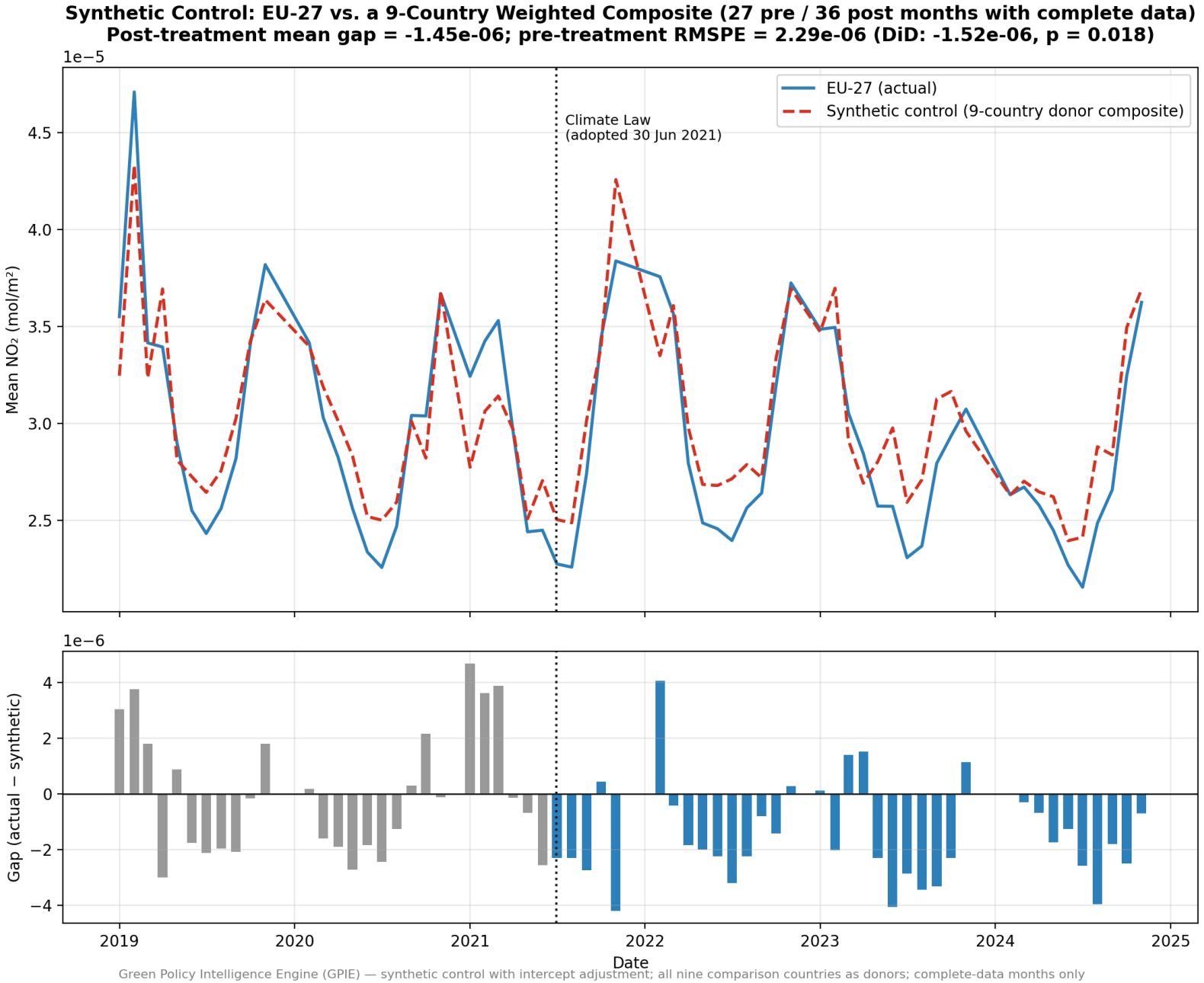
India Transferability Test

NO2 monthly means over India's national boundary (GADM 4.1), 2019-2024, using the EU-27 request builder. Acquisition test only, not a causal analysis.



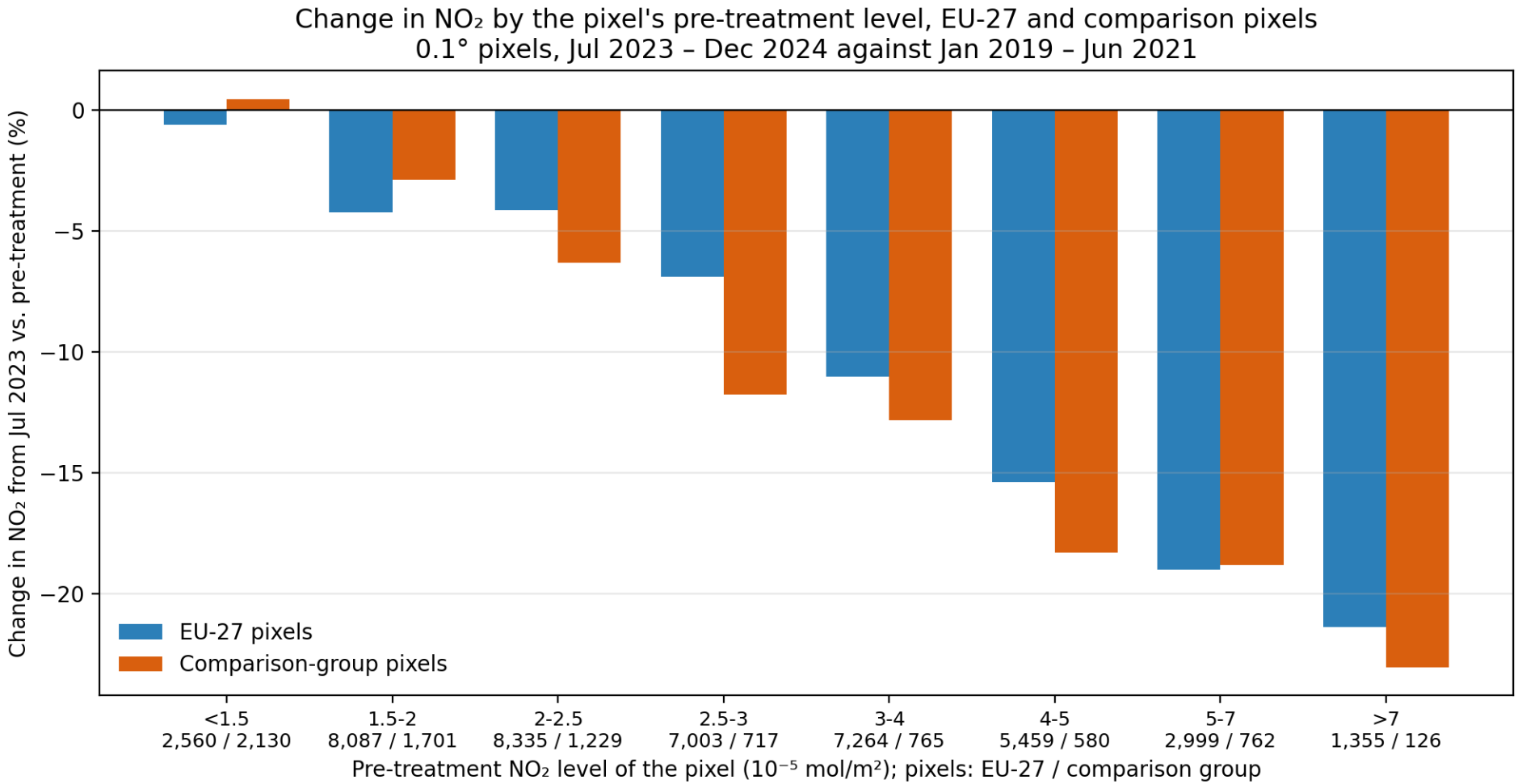
Synthetic Control: EU-27 vs. 9-Country Donor Composite

Convex donor weights with pre-period intercept adjustment; complete-data months only. Gap, pre-period RMSPE and DiD estimate in the figure title. All nine comparison countries are donors.



NO2 Change by Pre-Treatment Pollution Level

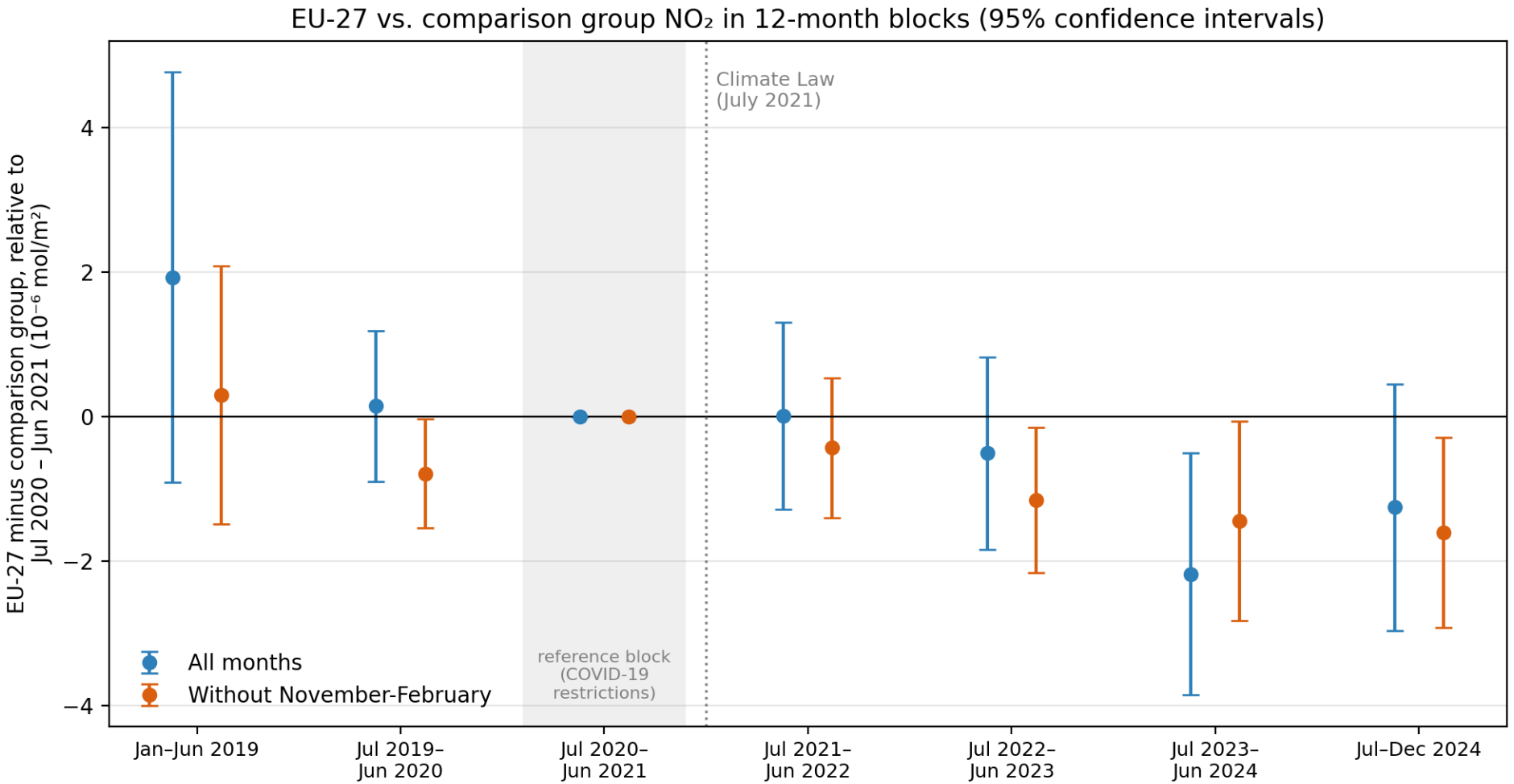
Change from July 2023 against the pre-treatment period, by the pixel's pre-treatment NO2 level, for EU-27 and comparison-group pixels (0.1 degree grid).



Source: Sentinel-5P TROPOMI tropospheric NO₂ column

EU-27 vs. Comparison Group in 12-Month Blocks

EU-27 minus comparison-group NO2 in 12-month blocks relative to July 2020 - June 2021, with and without November-February (95% confidence intervals).

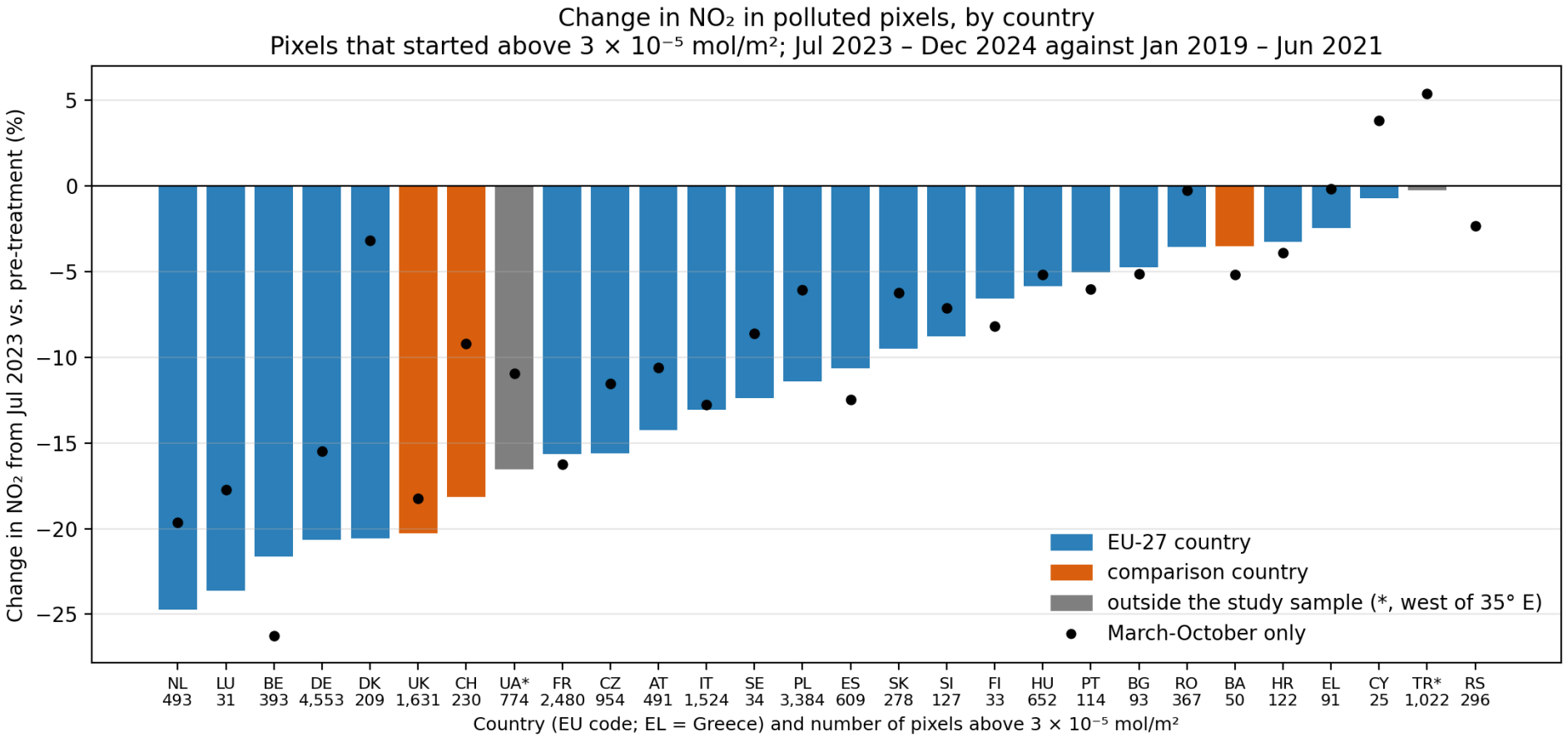


Source: Sentinel-5P TROPOMI tropospheric NO_2 column

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NO2 Change in Polluted Pixels by Country

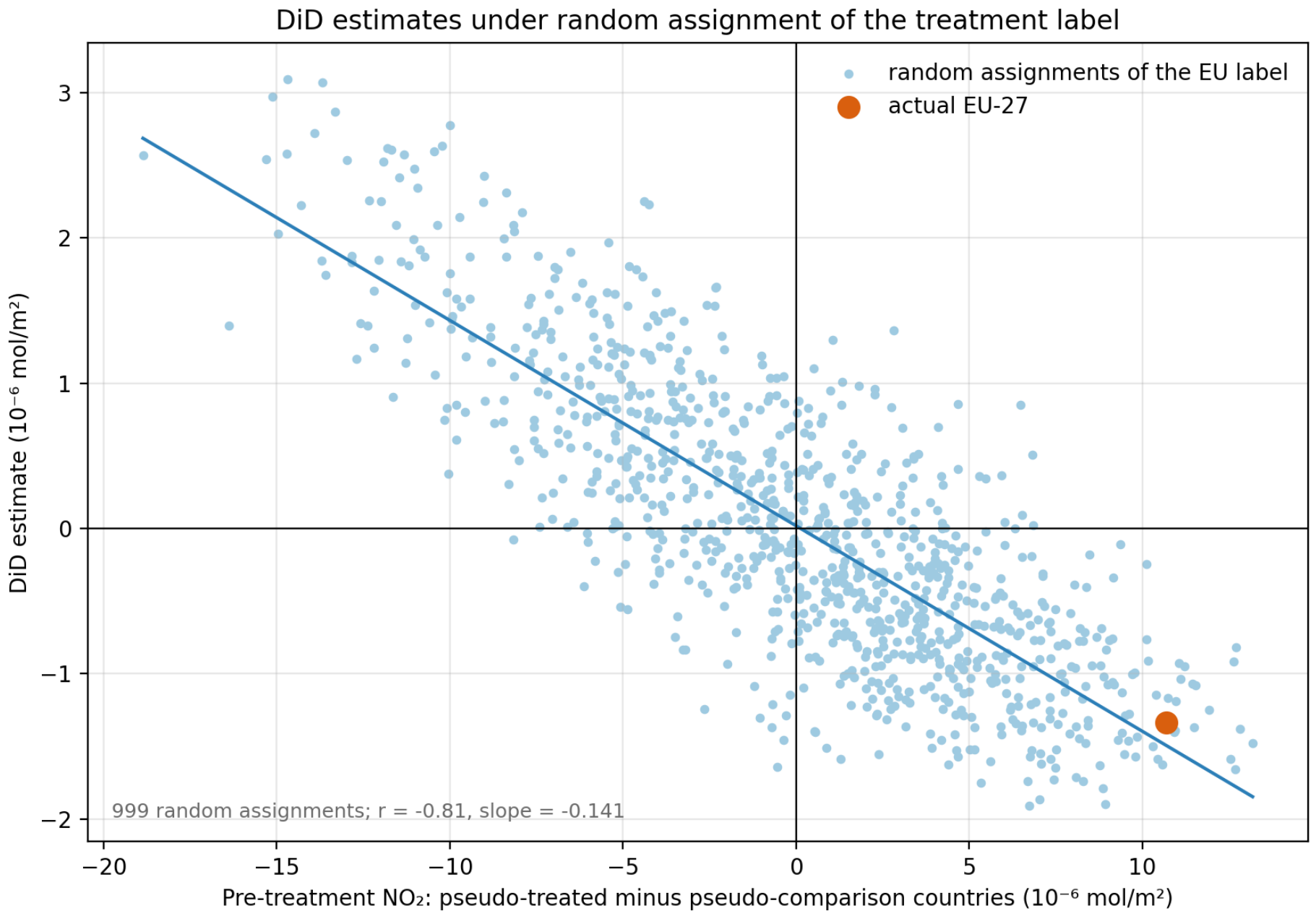
Change from July 2023 in pixels that started above 3e-5 mol/m2, by country. Dots: March-October only. Grey bars are areas outside the study sample.



Source: Sentinel-5P TROPOMI tropospheric NO_2 column

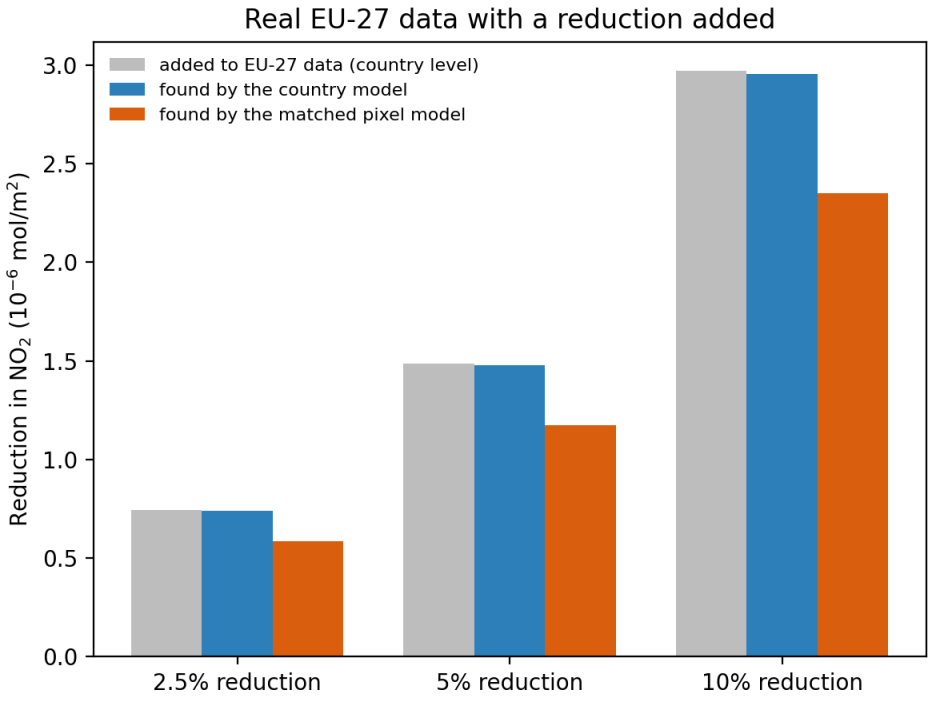
DiD Estimates under Random Group Assignment

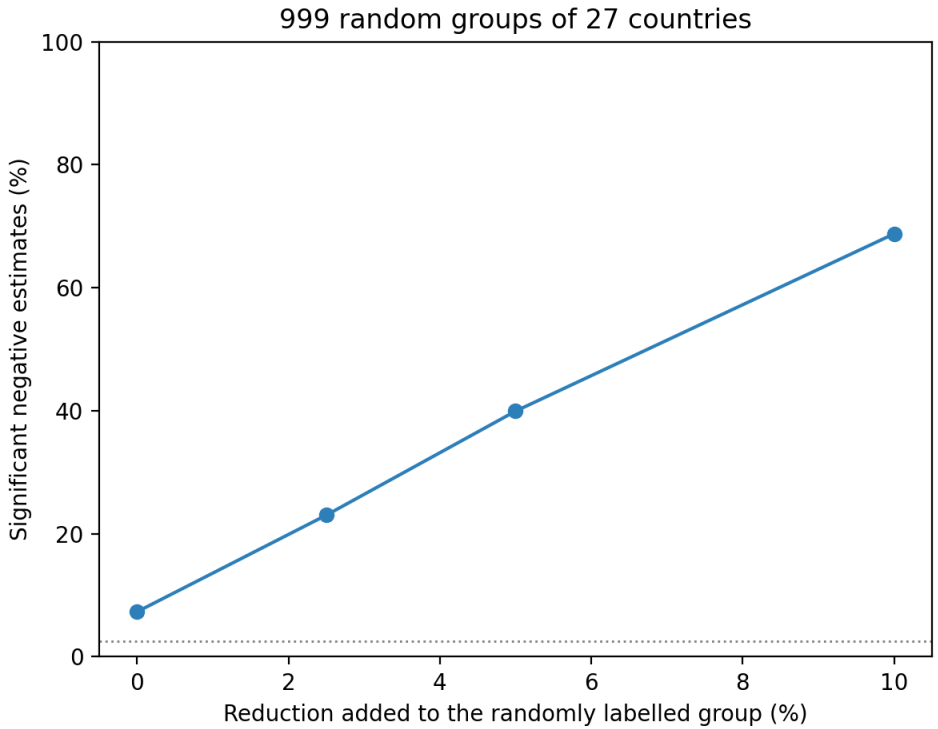
DiD estimate against the pre-treatment NO2 gap between groups, for random splits of the 36 countries into 27 and 9. The actual EU-27 assignment is marked.



Positive Control: Detecting a Known Reduction

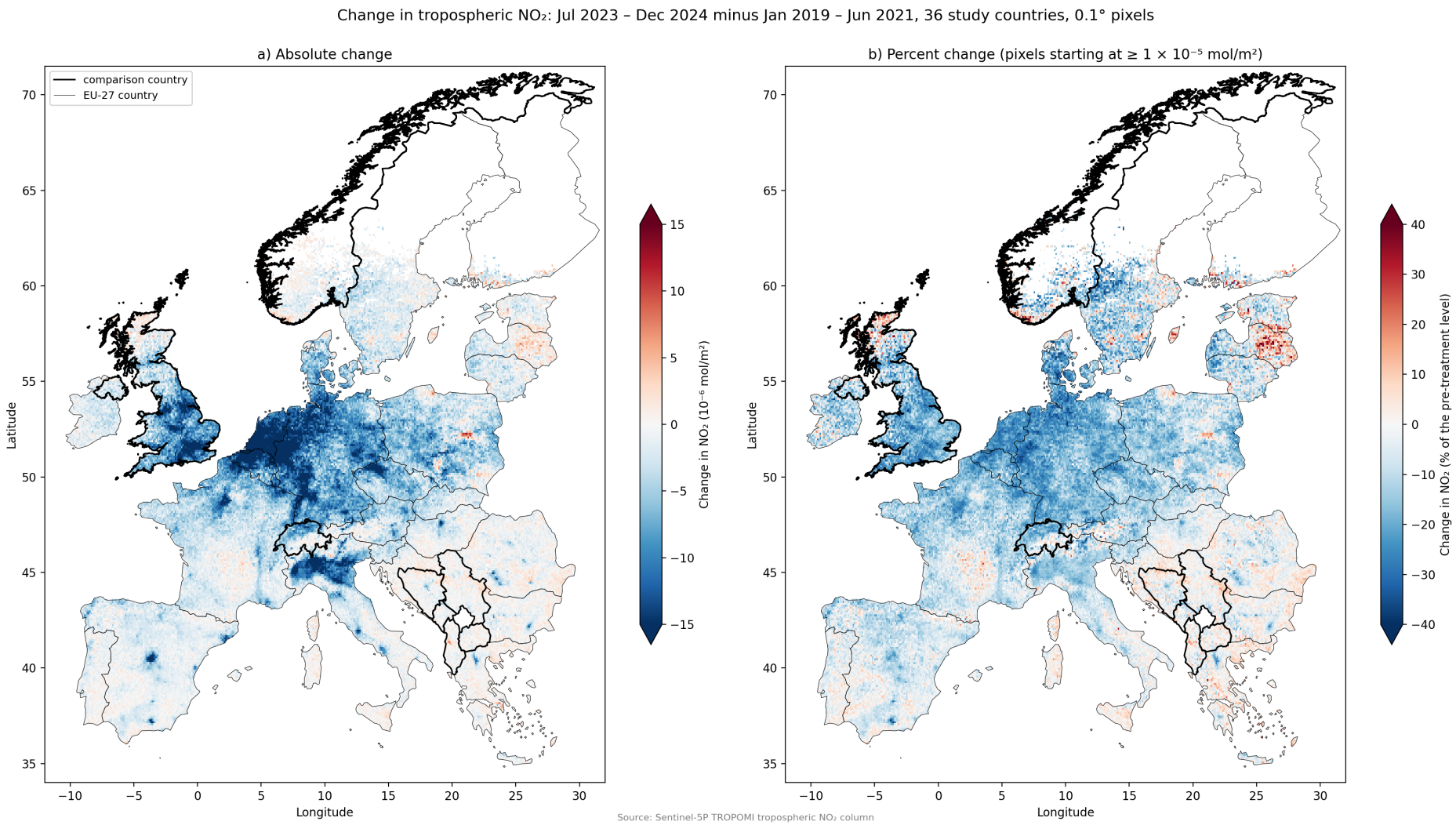
2.5%, 5% or 10% is added to EU-27 NO2 from July 2021. Left: the reduction added and what the country and matched pixel models find. Right: share of 999 random 27-country groups with a significa





Pixel-Level NO2 Change Map

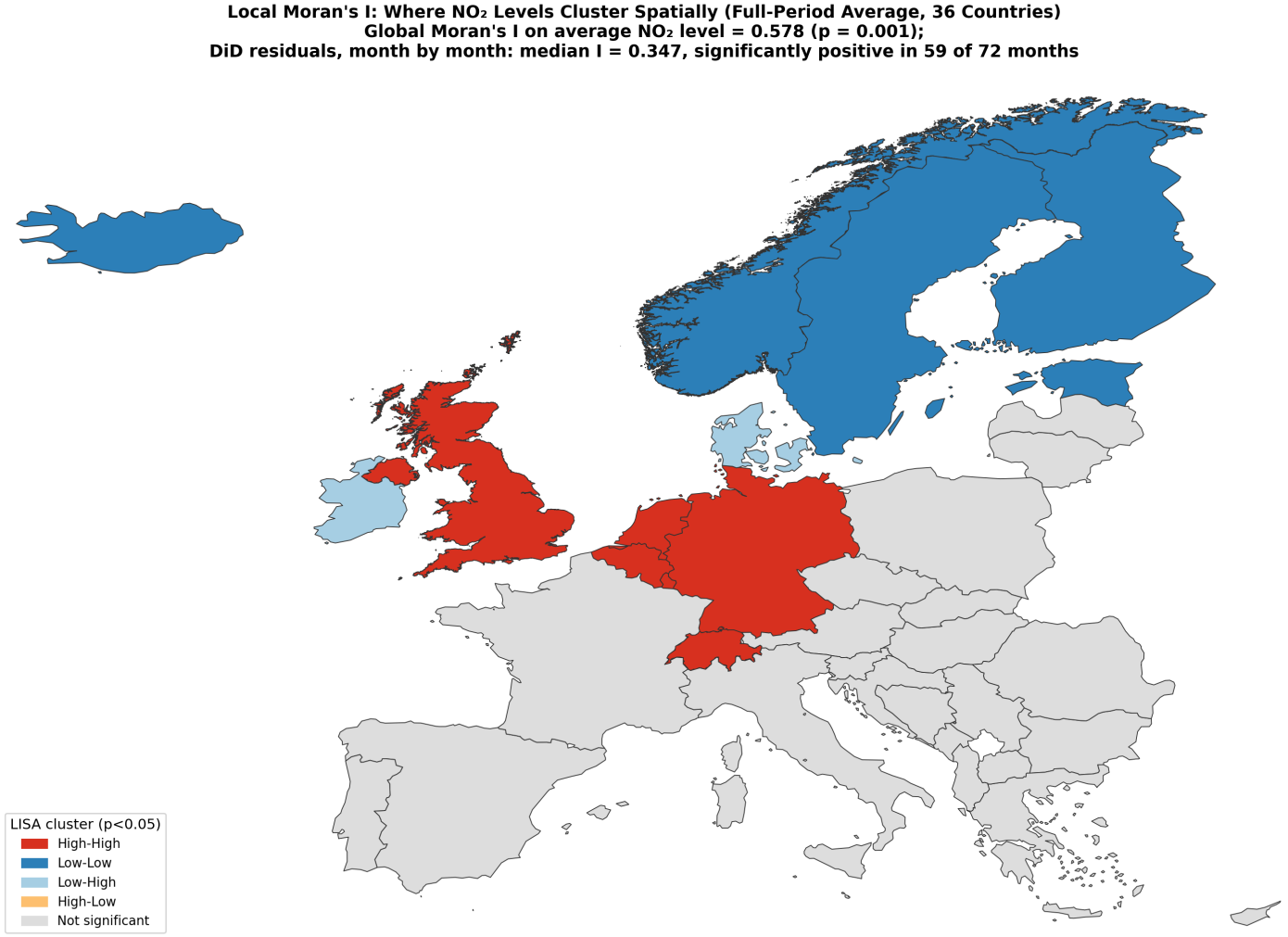
ange in tropospheric NO2 per 0.1 degree pixel, July 2023 - December 2024 minus January 2019 - June 2021: (a) absolute, (b) percent of the pre-treatment level for pixels starting at 1e-5 mol/m2 or mo



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Local Moran's I (LISA) Spatial Cluster Map

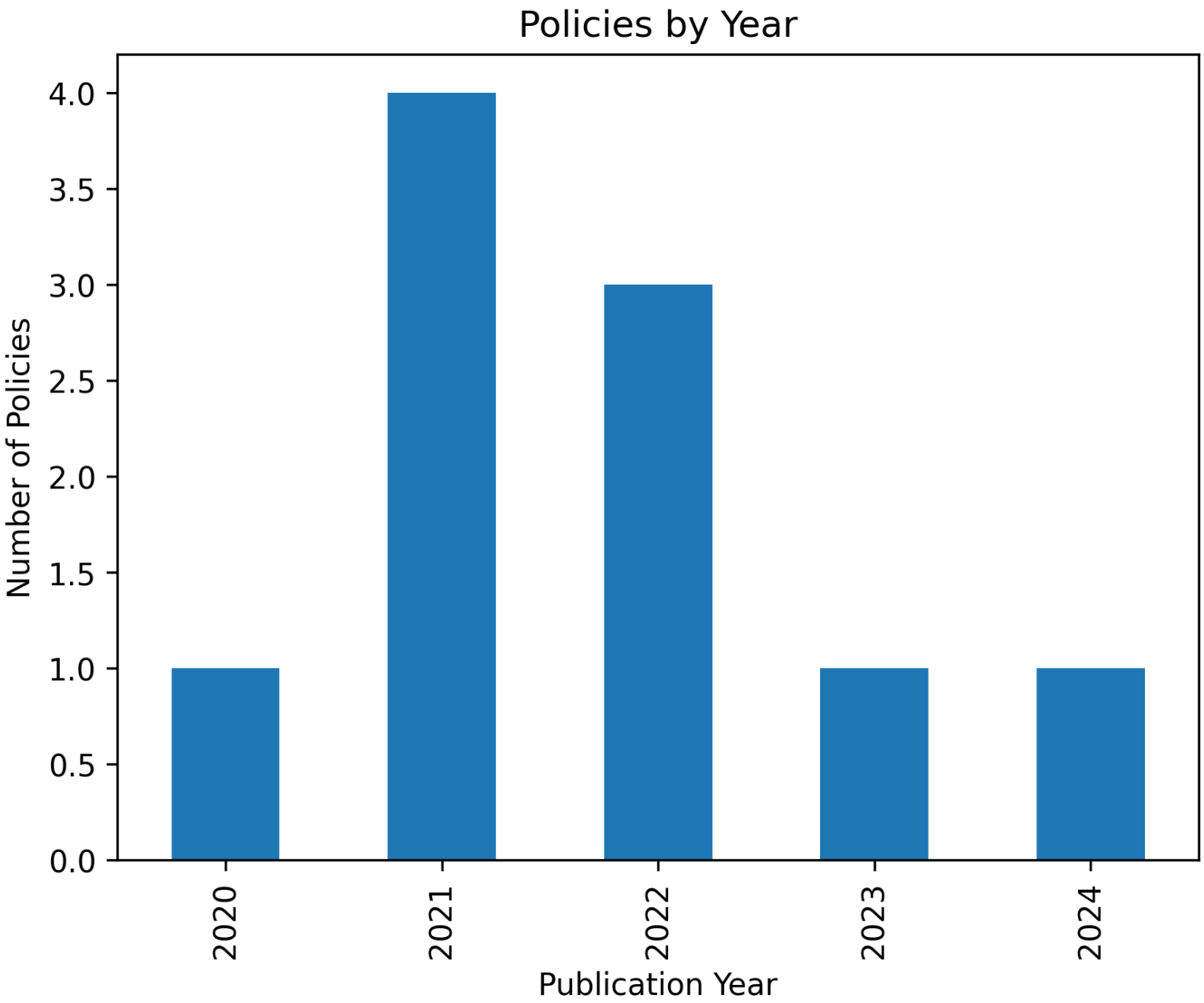
LISA clusters of full-period average NO2 (descriptive). Global and month-by-month residual Moran's I results in the figure title.



Green Policy Intelligence Engine (GPIE) — Local Moran's I (esda), KNN-4 spatial weights on country centroids

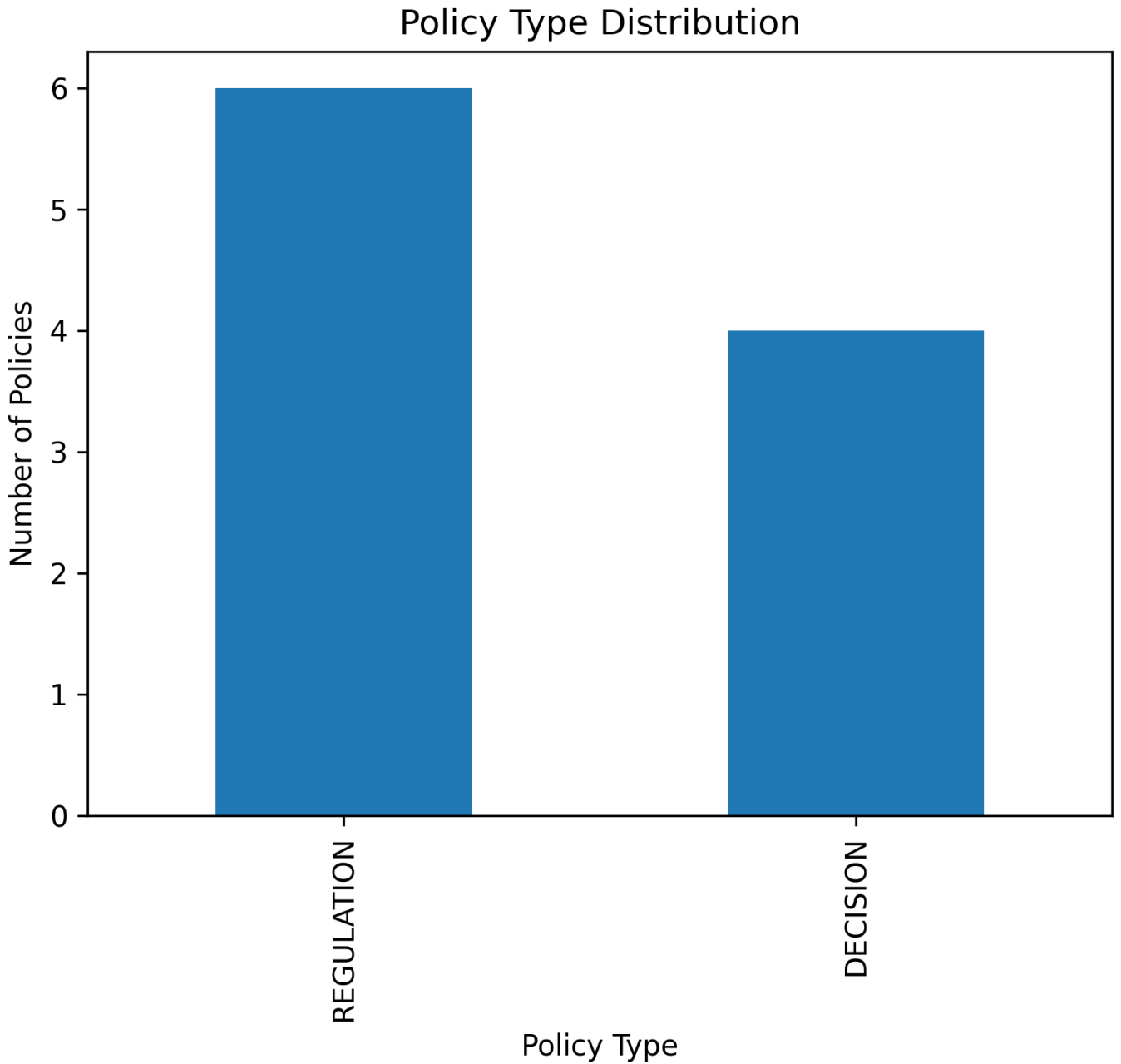
Policies by Year

Count of records per year in the 10-record EUR-Lex sample scraped for this project (not a complete policy inventory).



Policy Type Distribution

Breakdown of policy dataset by policy type.



Policy Types by Year

Policy type composition over time.

